

Interaksi Manusia dan Komputer (IMK)

THE USER INTERFACE (UI) DESIGN PROCESS

Step 2 : Understand The Business Function



Tujuan

Setelah mengikuti materi ini mahasiswa dapat:

1. Memahami pentingnya mengetahui fungsi bisnis berdasarkan *Persona user* sebagai acuan dalam merancang antarmuka perangkat lunak.
2. Mengerti dan menerapkan setiap tahapan dalam mengidentifikasi alur kerja setiap *task* berdasarkan *Persona User*.

General Step

1. Perform a business definition and requirements analysis.
2. Determine basic business functions.
3. Describe current activities through task analysis.
4. Develop a conceptual model of the system.
5. Establish design standards or style guides.
6. Establish system usability design goals.
7. Define training and documentation needs.

General Step (cont.)



1. Perform a business definition and requirements analysis

- The Objectives of this phase is to establish the need for a system.
- A product description is developed and refined, based on input from users, marketing, or other interested parties.
- The developer should be aware of the policies and work culture of the organization.
- Method of collection :
 - Data gathering is important part of the requirements discovery process in interaction design.
 - Data collection includes **observation** and **elicitation** methods.
 - **Observation methods** allow the designer to immerse themselves in user's environment in their day-to-day activity by watching them but users don't participate directly with the HCI designer.
 - **Elicitation methods** require user's participation and include direct and indirect methods such as interview, focus groups, and quistionnaire

Methods to Collect Required Data

- Direct Observation
- Questionnaire
- Interview
- Focus group discussion and brainstorming
- *Please explain what are their advantages and disadvantages respectively*
All of them can be performed using online media

- Observation is a requirements discovery technique wherein the HCI designers either participates in or watches a person perform activities to learn about the interaction.
- 3 minimum conditions before observation by Tull and Hawkins(1983):
 - The data has to be available for observation
 - The behavior has to be repetitive, frequent, or otherwise predictable
 - An event has to cover a reasonably short time span
- When observing in the requirements discovery, Whitten(2000) suggested the following point :
 - Determining the who, what, where, when, why, and how of the observation.
 - Obtain permission from appropriate supervisors or managers.
 - Inform those who will be observed of the purpose of the observation.
 - Keep a low profile.
 - Take notes during or immediately following the observation.
 - Review observation notes with individuals.
 - Don't interrupt the individuals at work.
 - Don't focus heavily on trivial activities.
 - Don't make assumptions.

Interview

- Interview are a requirements discovery technique whereby the interaction designer collects information from individuals through face-toface interaction.
- List of activities that could be used when preparing interviews (Whitten,2000):
 - Select interviewees.
 - Prepare for the interview with specific questions the interviewer will ask the interviewee.
 - Conduct the interview.
 - Follow up on the interview.

Focus Groups and Brainstorming

- **Focus groups**

- are process whereby highly structured group meetings are conducted for the purpose of analyzing problems and defining requirements.
- **Focus groups** require a facilitator role. Facilitator encourage user and management participation without allowing individuals to dominate the session.
- **Focus groups** actively involve users in interaction design and this improves their acceptance and reduces the risk of resistance at the implementation stage

- **Brainstorming**

- It is similar to focus group but more informal and use for generating ideas during group meetings.
- Participants are encouraged to generate as many ideas as possible in a short period of time without any analysis until all the ideas have been exhausted.

Questionnaires

- Questionnaires are special-purpose documents that allow the interaction designer to collect information opinions from respondents.
- Questionnaires can be in a free or fixed format.
- When performing data collection with the use of questionnaires (Whitten, 2000):
 - Determine what facts and opinions must be collected and from whom you should get them.
 - Based on the needed facts and opinions, determine whether free or fix format questions will produce the best answers.
 - Write the questions
 - Test the questions on a small sample respondents.
 - Duplicate and distribute the questionnaire.

Direct Methods

- ◆ Individual Face-to-Face interview
- ◆ Telephone Interview or Survey
- ◆ Traditional Focus Group
- ◆ Facilitated Team Workshop
- ◆ Observational Field Study
- ◆ Requirements Prototyping
- ◆ User-Interface Prototyping
- ◆ Usability Laboratory Testing
- ◆ Card Sorting for Web Sites



Indirect Methods

- ◆ MIS Intermediary
- ◆ Paper Survey or Questionnaire
- ◆ Electronic Survey or Questionnaire
- ◆ Electronic Focus Group
- ◆ Marketing and Sales
- ◆ Support Line



Indirect Methods (Continued)

- ◆ E-mail or Bulletin Board
- ◆ User Group
- ◆ Competitor Analyses
- ◆ Trade Show
- ◆ Other Media Analysis
- ◆ System Testing



An Example: Marjorie Bannet

- **Goals**

- To not be lonely
- Keep in touch with her sons and their families
- Avoid frustrating technology experiences!
- Not be reliant on anyone



2. Determine basic business functions

- A detailed description of what the product will do is prepared
- Major system functions are listed and described, including critical system inputs and outputs
- The process :
 - Gain a complete understanding of the user's mental model based upon
 - The user's needs and the user's profile
 - A user task analysis.
 - Develop a conceptual model of the system based upon the user's mental model, including :
 - Defining objects
 - Developing metaphors.

Mental Model

- Is an internal representation of a person's current conceptualization and understanding of something
- Mental models are gradually developed through experience, training, and instruction
- Enable a person to understand, explain, and do something
- Mental models enable a person to predict the actions necessary to do things if the actions have been forgotten or have not yet been encountered.

Understanding the User's Mental Model

- ◆ How users understand their task
- ◆ User's needs
- ◆ User's work preferences
- ◆ How they address the task
- ◆ Mental Models contain metaphors and objects
- ◆ Distill out these metaphors and objects to frame program

Ilustrasi:

<https://www.smashingmagazine.com/2012/04/mental-model-diagrams-cartoon/>



UX Knowledge Base Sketch #25

MENTAL MODEL



"IF AN ORGANISM CARRIES A 'SMALL-SCALE MODEL' OF EXTERNAL REALITY AND OF ITS POSSIBLE ACTIONS WITHIN ITS HEAD, IT IS ABLE TO TRY OUT VARIOUS ALTERNATIVES, CONCLUDE WHICH IS THE BEST OF THEM, REACT TO FUTURE SITUATIONS BEFORE THEY ARISE, UTILIZE THE KNOWLEDGE OF PAST EVENTS IN DEALING WITH THE PRESENT AND THE FUTURE."



INTERNAL
REPRESENTATION
OF



EXTERNAL
REALITY
BASED ON



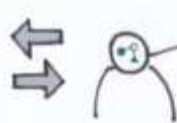
PAST EXPERIENCES,
LEARNING & INTUITION

⇒ "A MENTAL MODEL REPRESENTS A THOUGHT PROCESS FOR HOW SOMETHING WORKS." (SUSAN CAREY)

UX HOW DOES THIS APPLY TO UX DESIGN?



SOFTWARE/
APPLICATION



USER'S MENTAL MODEL
ABOUT HOW THIS WORKS
(WHICH IS IN CONSTANT CHANGE!)

EXPERIENCE WITH DIGITAL PRODUCTS IN GENERAL
EXPERIENCE WITH SIMILAR APPS
INTUITION
WHAT THE USER HEARD/READ ABOUT IT
+ EXPERIENCE WITH THIS GIVEN APP (OBVIOUSLY, THIS ONLY APPLIES TO RETURNING USERS)



THE CONCEPTUAL MODEL
SHOULD MATCH WITH
THE USERS' MENTAL MODELS



THERE CAN BE DIFFERENT
USER GROUPS WITH DIFFERENT
MENTAL MODELS
AS A RESULT,
A FAMILIAR MODEL
IMPROVES THE OVERALL
UX (E.G.: LEARNABILITY,
EFFICIENCY, MEMORABILITY,
USER SATISFACTION)

CONCEPTUAL MODEL:

"THE ACTUAL MODEL THAT IS GIVEN TO THE PERSON THROUGH THE DESIGN AND INTERFACE OF THE ACTUAL PRODUCT" (SUSAN M. WEINSCHENK)



HOW? TO GAIN INFORMATION ABOUT THE USERS' MENTAL MODELS?

- CONDUCT UX RESEARCH, →
LISTEN & OBSERVE



CONTEXTUAL INQUIRY

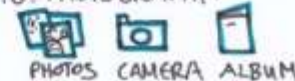
(E.G. WHAT ARE THE USER'S
EXPECTATIONS?
PRIOR KNOWLEDGE/
EXPERIENCE?
HOW IT SHOULD WORK?)



USER INTERVIEW

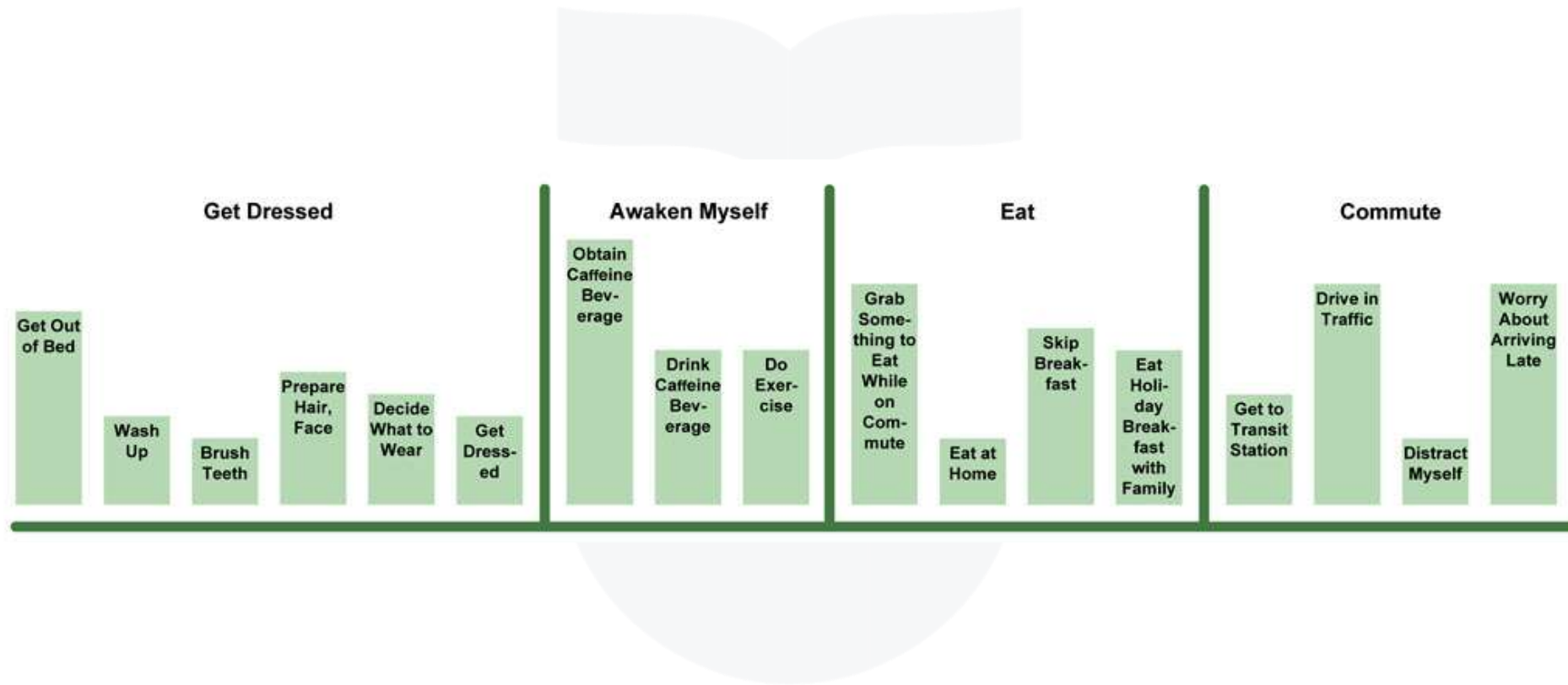
- IDENTIFY METAPHORS:
FAMILIAR CONCEPTS HELP THE USERS SETTING THEIR
EXPECTATIONS - E.G.: DESKTOP METAPHOR; FOLDER
- PAY ATTENTION TO THE TERMINOLOGY THE USERS APPLY.
- IDENTIFY THE OBJECTS, CONCEPTS RELATED TO
A TASK / ACTIVITY, E.G. PHOTOGRAPHY:

THESE ARE
THE COMPONENTS:



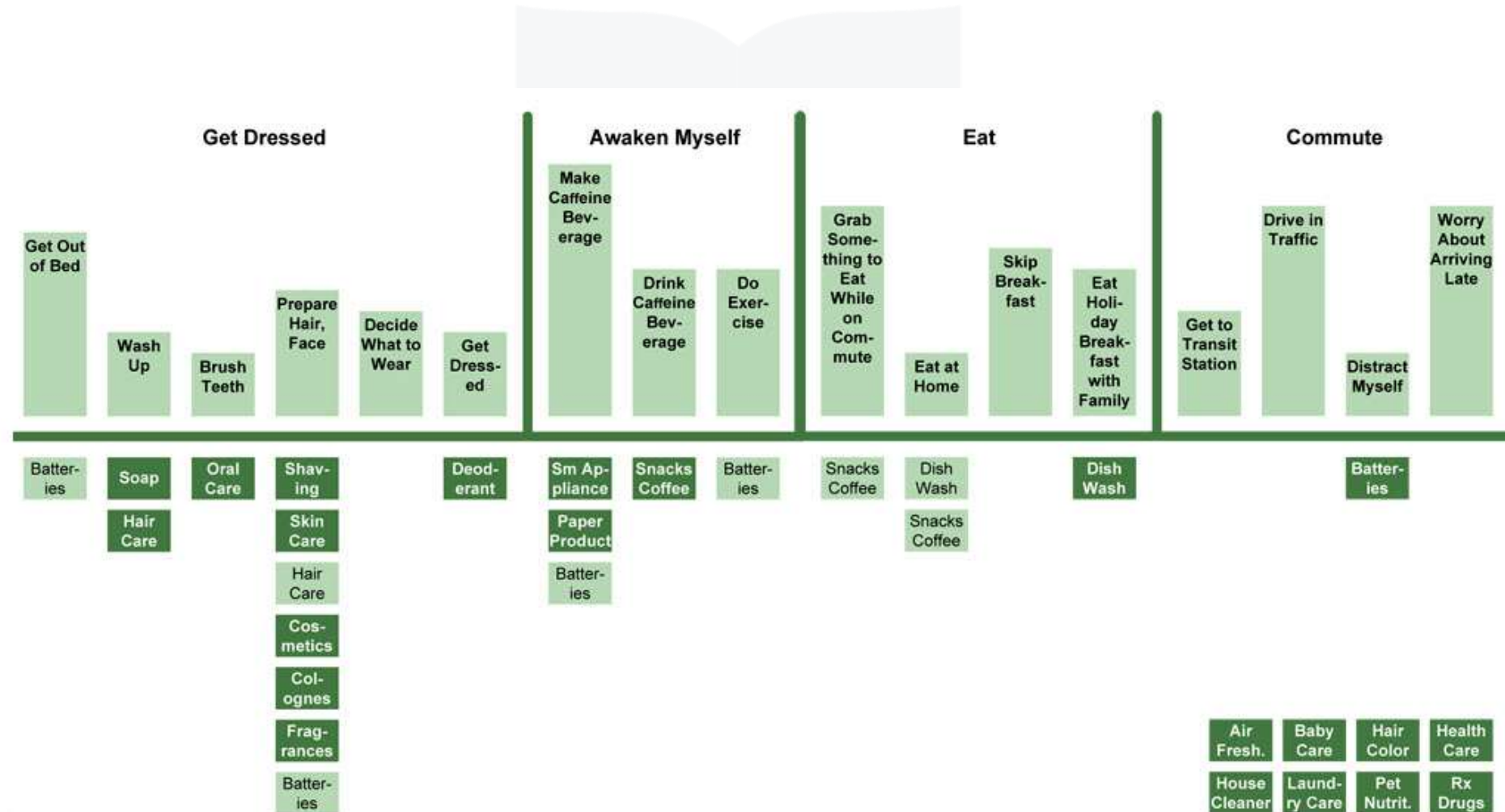
⇒ THE CONCEPTUAL MODEL SHOULD REFLECT THESE

Contoh 1

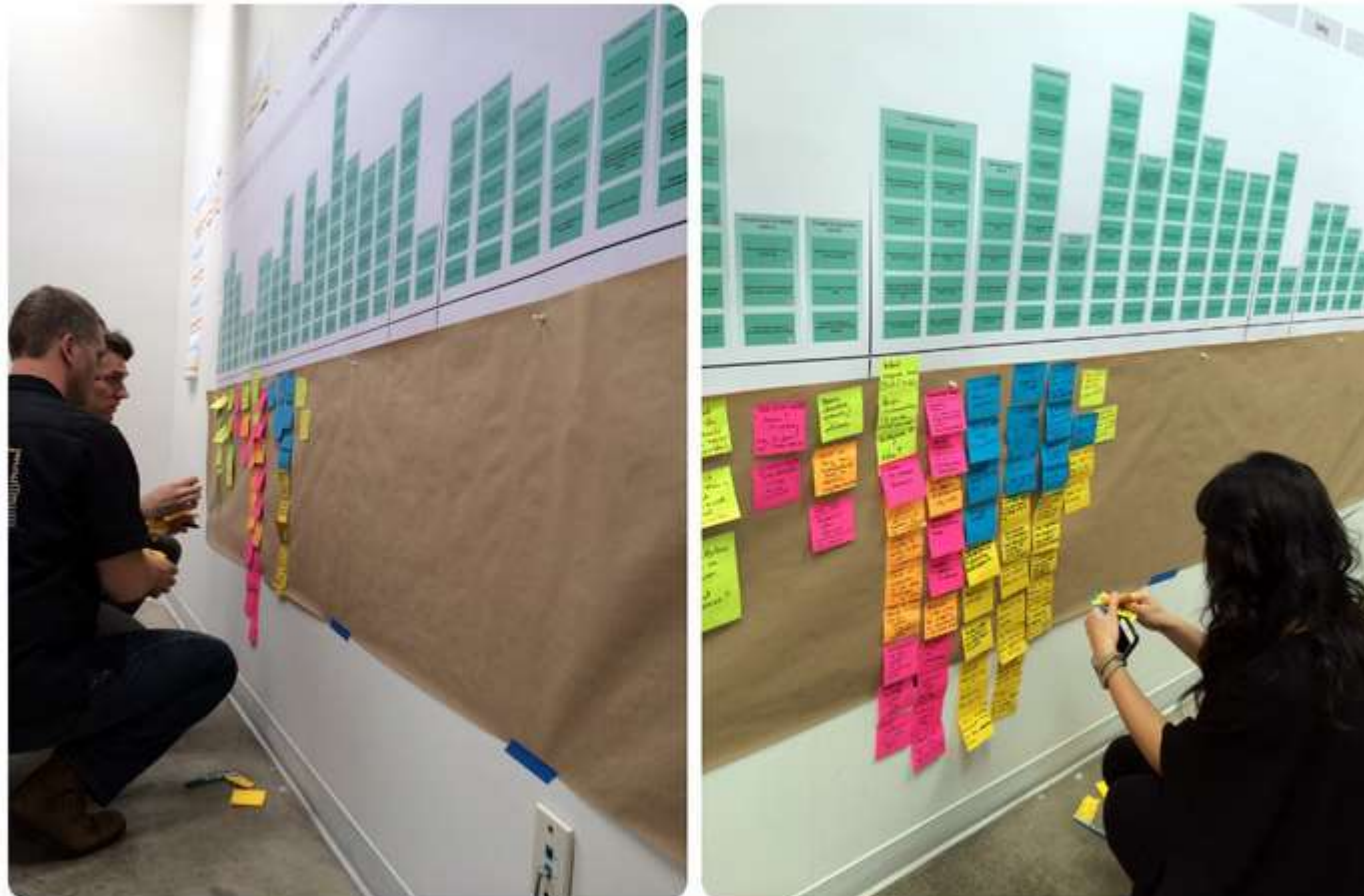


<https://www.crazyegg.com/blog/how-to-design-mental-models/>

Contoh 2



<https://medium.com/@workatplay/how-to-create-an-audience-mental-model-fb5ee03a7c7f>



3. Task Analysis

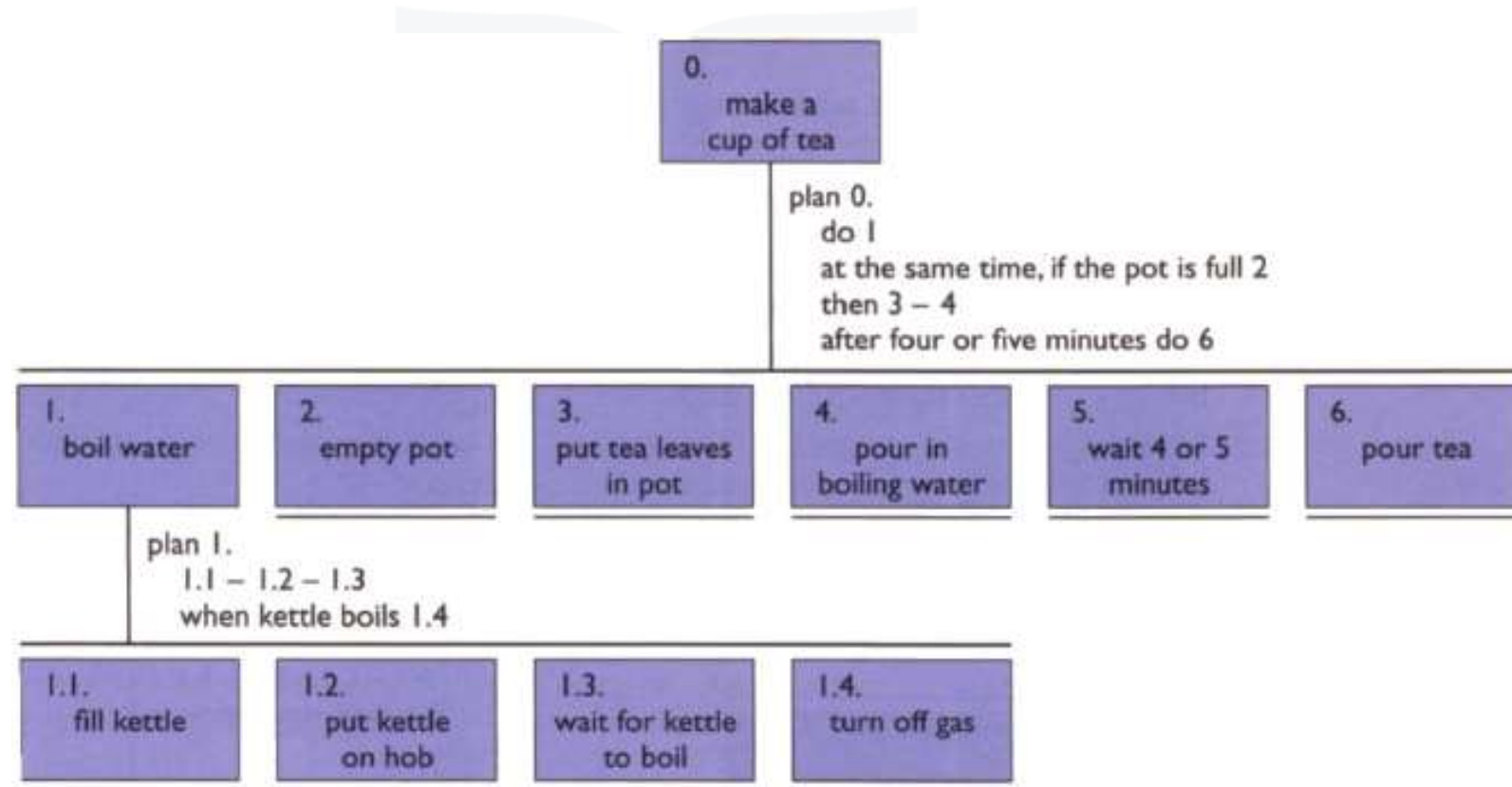
- Task Analysis is a set activities that change the systems from initial state to a specified goal or desired outcome state.
- The outcome may involve a significant change to the current state, so we split tasks into a sequence of subtask, each more simple than the parent task.
- Rees(2001) propped a list of activities in order to conduct task analysis and are described below :
 - Gathering information from observation of and/or consulting with user.
 - Representing tasks in a task description notation
 - Performing an analysis of the task descriptions to achieve an optimum description
 - Using the task representation to produce a new user interface design or improve an existing one
- Other tools : ishikawa diagram, Use cases, Story boarding, Primary stakeholder profiles.

Hierarchical Task Analysis (HTA)

- The most popular technique used for this type of analysis is the Hierarchical Task Analysis (HTA) tool.

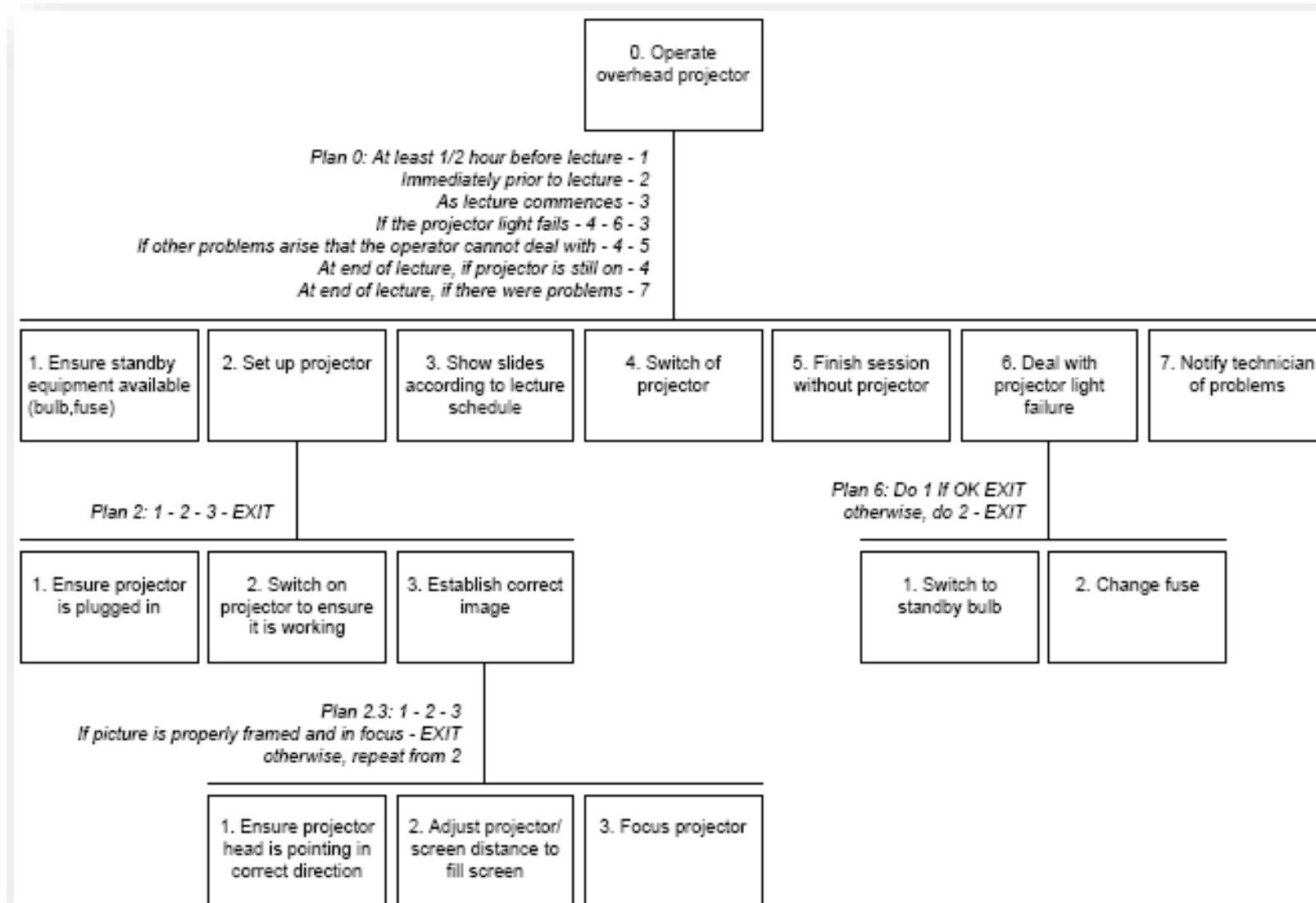


Content HT A



[Dix et al, p. 515]

Contoh HTA





Bahan Bacaan

Hierarchical Task Analysis (HTA)

1. <https://www.uxmatters.com/mt/archives/2010/02/hierarchical-task-analysis.php>
2. http://cehdclass.gmu.edu/ndabbagh/Resources/IDKB/hierarchical_analysis.htm
3. <http://www.taskarchitect.com/blog/what-is-hierarchical-task-analysis>
4. <http://www.taskarchitect.com/blog/when-should-i-use-task-analysis>
5. <http://www.taskarchitect.com/blog/when-to-stop-breaking-down-task-in-hierarchical-task-analysis>
6. <http://disi.unitn.it/~deangeli/homepage/lib/exe/fetch.php?media=teaching:bolzano:taskanalysis.pdf>

The 7 Steps of Hierarchical Task Analysis

1. Part 1 → <http://www.taskarchitect.com/blog/7-steps-hierarchical-task-analysis>
2. Part 2 → <http://www.taskarchitect.com/blog/the-7-steps-of-hierarchical-task-analysis-part-2>
3. Part 3 → <http://www.taskarchitect.com/blog/the-7-steps-of-hierarchical-task-analysis-part-3>
4. Part 4 → <http://www.taskarchitect.com/blog/the-7-steps-of-hierarchical-task-analysis-part-3-0>
5. Part 5 → <http://www.taskarchitect.com/blog/the-7-steps-of-hierarchical-task-analysis-part-3-0-0>
6. Part 6 → <http://www.taskarchitect.com/blog/hierarchical-task-analysis-part-6>
7. Part 7 → <http://www.taskarchitect.com/blog/7-steps-of-hierarchical-task-analysis-part-7>
8. Final → <http://www.taskarchitect.com/blog/7-steps-of-hierarchical-task-analysis-final>

CHALLENGE

- Create a HTA by your group
 - Online ojek application, consisted of booking/reservation until rating system
 - Graveyard information system, consisted of booking, payment, until post-sales service
 - Driving licences examination, consisted of registration until issuing

4. Developing a conceptual model

- Is the general conceptual framework through which the system's functions are presented.
- Describes how the interface will present objects, the relationships between objects, the properties of objects, and the actions that will be performed.
- A conceptual model is based on the user's mental model.
- The goal : to facilitate for the user the development of a useful mental model of the system.
- Several tools for conceptual design : brainstorming, card sort, personas, scenarios, semantic networks and cognitive walkthroughs.

Guidelines Designing Conceptual Models

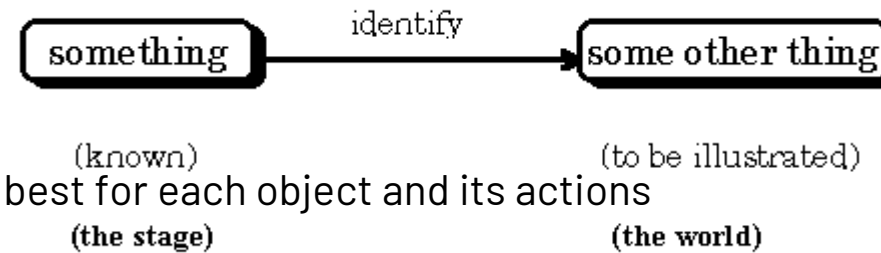
- Reflect the user's mental model, not the designer's
- Draw physical analogies or present metaphors
- Comply with expectancies, habits, routines, and stereotypes
- Provide action-response compatibility
- Make invisible parts and processes of a system visible
- Provide proper and correct feedback
- Avoid anything unnecessary or irrelevant
- Provide design consistency
- Provide documentation and a help system that will reinforce the conceptual model
- Promote the development of both novice and expert mental models

Defining Objects

- Determine all objects that have to be manipulated to get work done. Describe
 - The objects used in tasks.
 - Object behavior and characteristics
 - The relationship of objects
 - The actions performed
 - The objects to which actions apply
 - Information or attributes that each object in the task must preserve, display, or allow to be edited
- Identify the objects and actions that appear most often in the workflow
- Make the several most important objects very obvious and easy to manipulate

Developing Metaphors

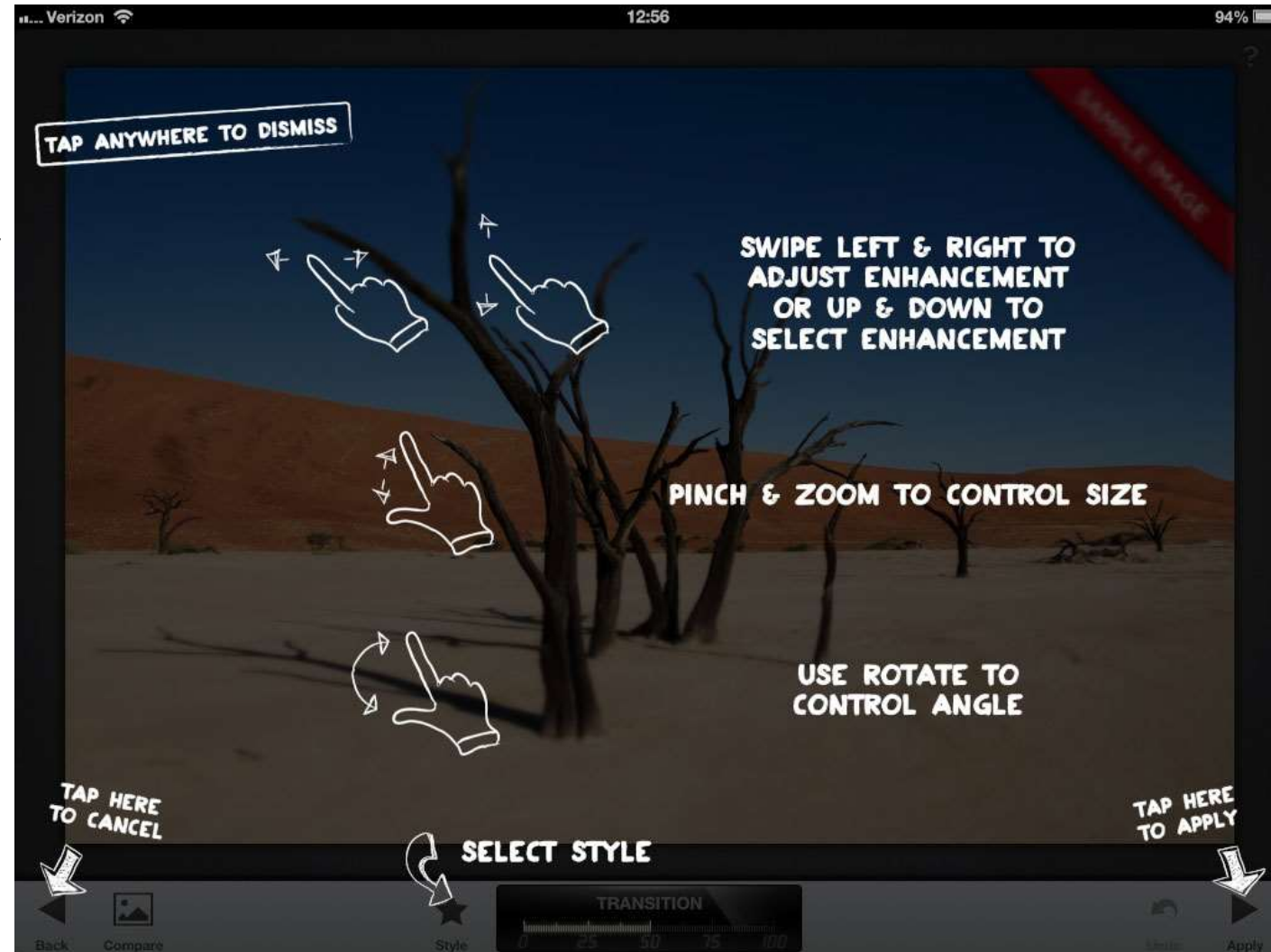
- A metaphor is a concept where one's body of knowledge about one thing is used to understand something else



- Guidelines :
 - Choose the analogy that works best for each object and its actions
 - Use real-world metaphors
 - Use simple metaphors
 - Use common metaphors
 - Multiple metaphors may coexist
 - Use major metaphors, even if you can't exactly replicate them visually
 - Test the selected metaphors

Example

<http://asktog.com/atc/principles-of-interaction-design/>

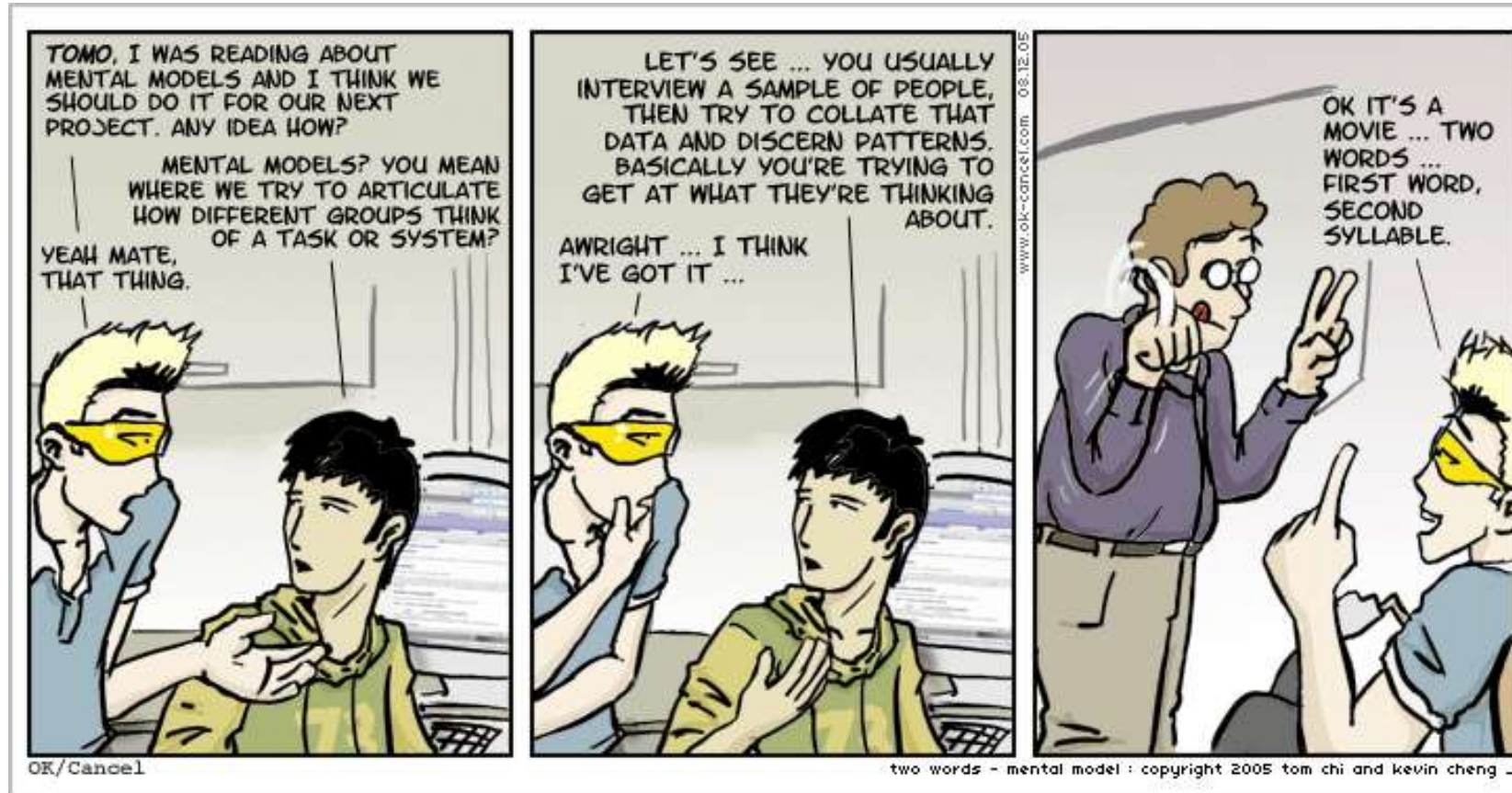


Metaphor Supplementary

- <http://www.katalinszabo.com/metaphor.htm>
- http://ecog.info/gerv/Resume/MS_in_IT_files/COMM6960Assignment2.pdf
- http://www.diku.dk/OLD/undervisning/2003f/multimedie/modul4/MOT_Evaluation.pdf
- <http://courses.cs.washington.edu/courses/cse595/98wi/metaphor.html>
- <http://www.ics.uci.edu/~kobsa/courses/ICS104/course-notes/metaphors.htm>

The User's New Mental Model

- When the system is implemented, a person interacts with the new system and its interface.
 - If the designer has correctly reflected the user's mental model in design: the user's mental model is reinforced and a feeling that the interface is intuitive will likely develop
 - if the new system does not accurately reflect the user's existing mental model : the results include breakdowns in understanding, confusion, errors, loss of trust, and frustration → an inability to perform the task or job
- If there was a gap between conceptual model and mental model, then tried to bridge this gap through extensive documentation and training.



Mental Model Supplementary:

<http://okcancel.com/archives/article/2005/08/mental-model-building-step-by-step.html>

5. Establish design standards or style guides.

- In interface design it describes the appearance and behavior of the interface and provides some guidance on the proper use of system components.
- It also defines the interface principles, rules, guidelines, and conventions that must be followed in detailed design
- It will be based on the characteristics of the system's hardware and software, the principles of good interface and screen design

Value of Standards and Guidelines

- To Users :
 - Allow faster performance
 - Reduce errors
 - Reduce training time
 - Foster better system utilization
 - Improve satisfaction
 - Improve system acceptance
 - Reduce development and support costs
- To System Developers :
 - Increase visibility of the human-computer interface
 - Simplify design
 - Provide more programming and design aids, reducing programming time
 - Reduce redundant effort
 - Reduce training time
 - Provide a benchmark for quality control testing

- 1981 : The first text to present a compilation of interface and screen design guidelines was that of Galitz (1981).
- 1986 : The United States Air Force released a set of design guidelines for its user interface designers

Commercial Style Guides
Apple Human Interface Guidelines for the Macintosh http://developer.apple.com/documentation/UserExperience/Conceptual/OSXHIGuidelines
IBM Ease of Use Web site www-03.ibm.com/easy/page/558
Microsoft Windows XP User Interface Guidelines http://www.microsoft.com
Sun Microsystems Java Look and Feel Design Guidelines http://java.sun.com/products/jlf/ed2/book/HIGTitle.html

UI Standard

- A user interface standard is an official set of internationally agreed-upon design approaches and principles for interface design.
 - ISO 9241: Ergonomic requirements for office work with visual display terminals
 - ISO 14915: Software ergonomics for multimedia user interfaces
 - ISO 13407: Human-centered design processes for interactive systems
 - ISO/CD 20282: Ease of operation of everyday products

Web Guidelines and Style Guides

- There has not been an opportunity for conventions and style guides to be properly developed and then accepted by the development community
- De facto standards are being established when an overwhelming majority of big sites focus on one way to do something
- Worldwide standards are also being looked at by organizations such as the World Wide Web Consortium (2001)

Customized Style Guides

- Include checklists to present principles and guidelines.
- Provide a rationale for why the particular guidelines should be used.
- Provide a rationale describing the conditions under which various design alternatives are appropriate.
- Include concrete examples of correct design.
- Design the guideline document following recognized principles for good document design.
- Provide good access mechanisms such as a thorough index, a table of contents, glossaries, and checklists.

6. Establish system usability design goals.

- Use all available reference sources in creating the guidelines.
- Use development and implementation tools that support the guidelines.
- Begin applying the guidelines immediately

7. System Training and Documentation Needs

- **Training**

- System training will be based on user needs, system conceptual design, system learning goals, and system performance goals.
- Training may include such tools as formal or video training, manuals, online tutorials, reference manuals, quick reference guides, and online help.

- **Documentation**

- System documentation is a reference point, a form of communication, and a more concrete design – words that can be seen and understood.
- It will also be based on user needs, system conceptual design, and system performance goals
- Creating documentation during development progress will uncover issues and reveal omissions that might not otherwise be detected until later in the design process.
- Any potential problems can be identified and addressed earlier in the design process, again reducing later problems and modification costs.

- ◆ Training is necessary for most programs
- ◆ Documentation is important



Interaksi Manusia dan Komputer (IMK)

THE USER INTERFACE (UI) DESIGN PROCESS

Step 3 : Understand the Principles of Good Interface and Screen Design



Tujuan

Setelah mengikuti materi ini mahasiswa dapat:

1. Memahami prinsip design interaksi.
2. Menerapkan prinsip design interaksi dalam merancang antarmuka perangkat lunak untuk menghasilkan antarmuka perangkat lunak yang baik.

A well-designed interface and screen

- Mencerminkan kemampuan, kebutuhan, dan tugas penggunaanya. Dikembangkan dalam batasan fisik yang dipaksakan ditampilkan oleh perangkat keras.
- Memanfaatkan kemampuan perangkat lunak pengendalian secara efektif.
- Mencapai tujuan bisnis dari sistem yang dirancang.

Human Considerations in Interface and Screen Design

- **Cara Mencegah Timbul Masalah dari Pengguna**

Semua gangguan dan hal yang menyebabkan putus asa harus dihilangkan dalam desain.

- **Apa yang Pengguna Inginkan**

Arah yang diinginkan adalah menuju **kesederhanaan, kejelasan, dan dimengerti**

- **Apa Pengguna Lakukan**

Ketika berinteraksi dengan komputer, pengguna:

- Mengidentifikasi tugas yang harus dilakukan atau kebutuhan yang harus dicapai
- Memutuskan bagaimana tugas akan diselesaikan atau kebutuhan terpenuhi.
- Komputer Memanipulasi kontrol.
- Mengumpulkan data yang diperlukan atau konten sementara menyaring data yang bermakna.
- Bentuk penilaian menghasilkan keputusan yang relevan dengan tugas atau kebutuhan.

Interface Design Goals

- Reduce visual work.
- Reduce intellectual work.
- Reduce memory work.
- Reduce motor work.
- Minimize or eliminate any burdens or instructions imposed by technology.

Three basic design principles (Rees,2001)

1. Consistency

- Is property that **makes an interaction predictable** from the user's perspective.
- Two basic rules consistency :
 - First, the interface must show consistency in the label used for it.
 - Second, that all user actions must be reversible; in other words, everything must have an "undo" feature.

2. Simplicity

- **Show no more than is needed** to achieve the desired user goal
- Require a **minimum of input** from the user
- Keep both the user and the task in **focus**
- Make **important concepts** particularly **clear**
- Use visual representations with **direct manipulation** (Where possible)
- **Clarity** is important to achieve simplicity : information representation, organization and naming of interface control, flow of control (dynamic behavior), Screen layout

Three basic design principles (Rees,2001) Cont.

3. Context

Context refers to ability that the **interaction should appear as sequence identifiable and distinct interface views**.

Contextual visibility can be incorporated into an interaction with three major tools :

1. **What-you-see-is-what-you-get (WYSIWYG)** : technical slang for representing information on the screen in the identical form(within display constraint) as it will appear in other media(such as paper and audio/visual). This includes both text and images-drawing, pictures, and (given the technology) moving images and sound.
 2. **Properties**. Each visual object possesses a range of attributes(size, shape, color, and other parameters) that can be controlled by user. This property information appears on the screen in property boxes.
 3. **Dialogs**. Whenever small amounts of information(text, number, option, yes/no) must be solicited from the user, use a minor variation on the interface view known as a dialog box. This disappears when the input is achieved.
- And many other more sophisticated principles from Shneiderman, Mayhew, IBM.

User interface design guidelines

- Many of the guidelines refer to character-based screens, but include many guidelines for the GUI. The guidelines can be split into six sections that are still valid today :
 1. Data Entry
 2. Data display
 3. Sequence control
 4. User guidelines
 5. Data transfer
 6. Data protection
- Another important guideline used in industry is the GNOME (Human Interface Guidelines (HIG)) <http://library.gnome.org/devel/hig-book/stable/>) serta guidelines lain yang telah disampaikan pada pertemuan sebelumnya.

Bahan bacaan

- http://web.cs.wpi.edu/~matt/courses/cs563/talks/smartin/int_design.html
- <https://developer.gnome.org/hig/stable/>
- <http://www.ambysoft.com/essays/userInterfaceDesign.html>
- <http://bokardo.com/principles-of-user-interface-design/>



Fakultas Informatika
School of Computing
Telkom University



THANK YOU